

Exploring HCI

Lecture 2

Tim RMK GIGA

Departemen Teknik Informatika



Kampus
Merdeka
INDONESIA JAYA



DEPARTEMEN
TEKNIK INFORMATIKA

Institut Teknologi Sepuluh Nopember

Technology

Emerging Technology

Tim RMK GIGA
Departemen Teknik Informatika



Institut Teknologi Sepuluh Nopember



ITS_campus

its.ac.id

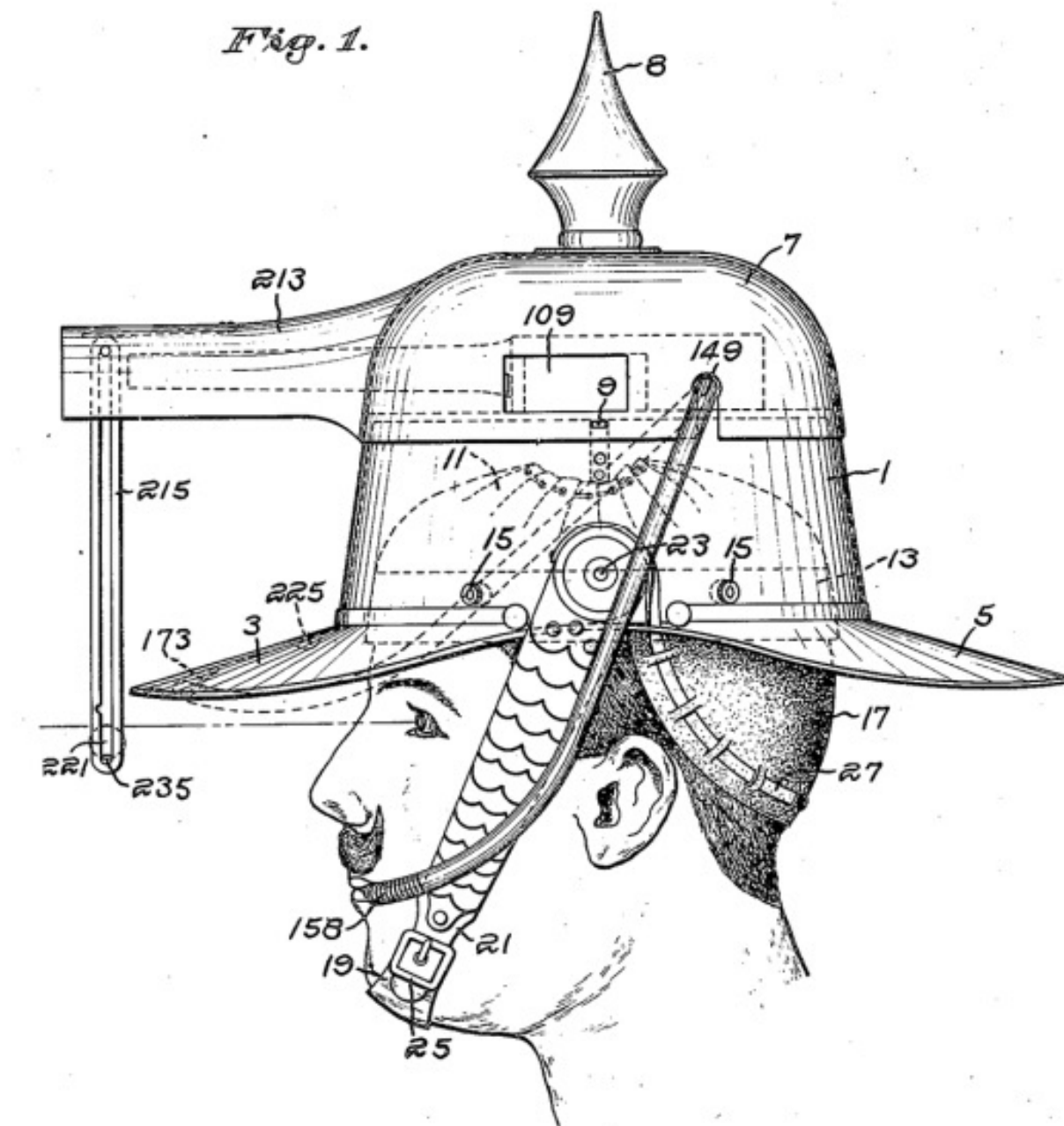
Sejarah VR



Stereoscope: 1832

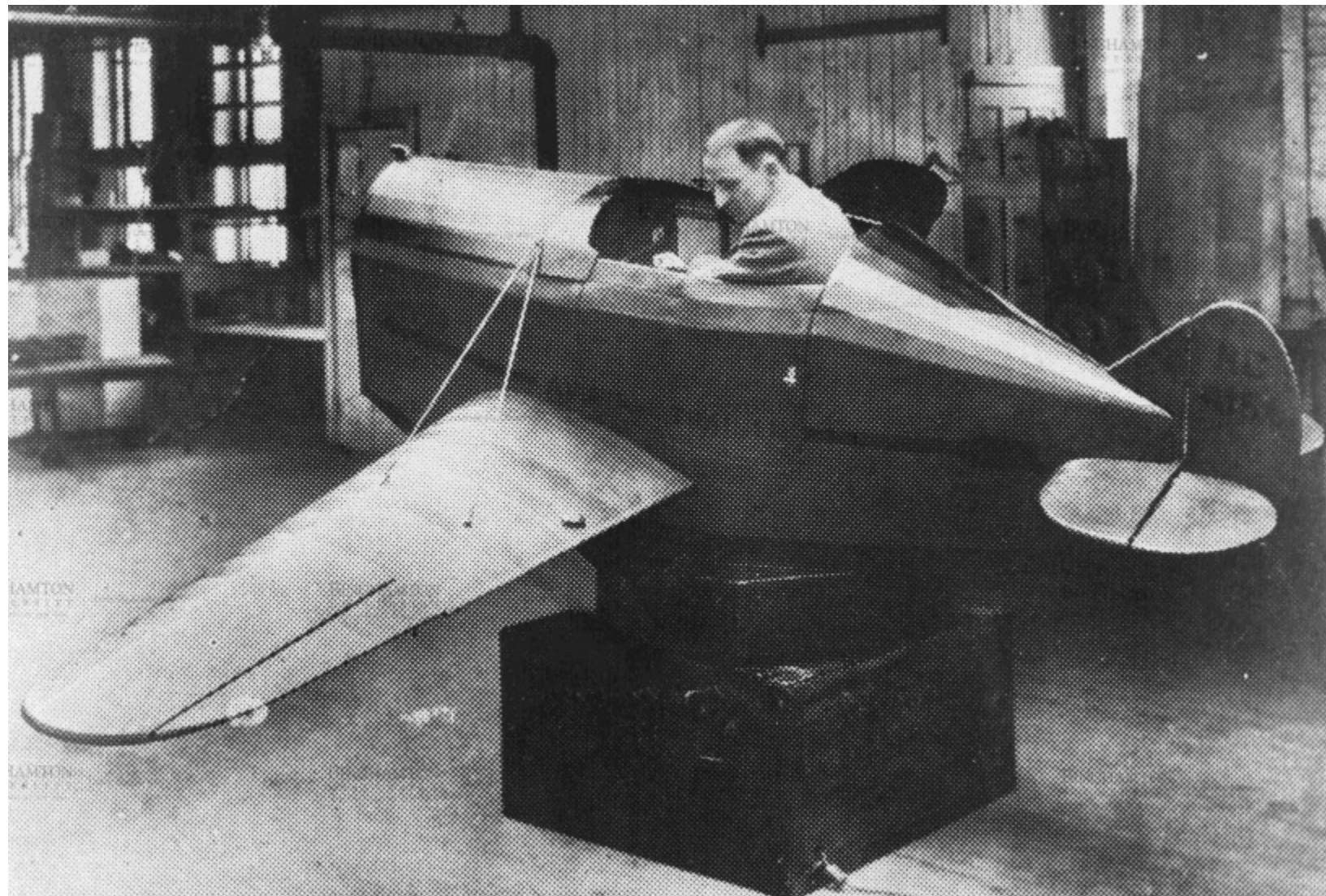


Hand free Stereoscope: 1851

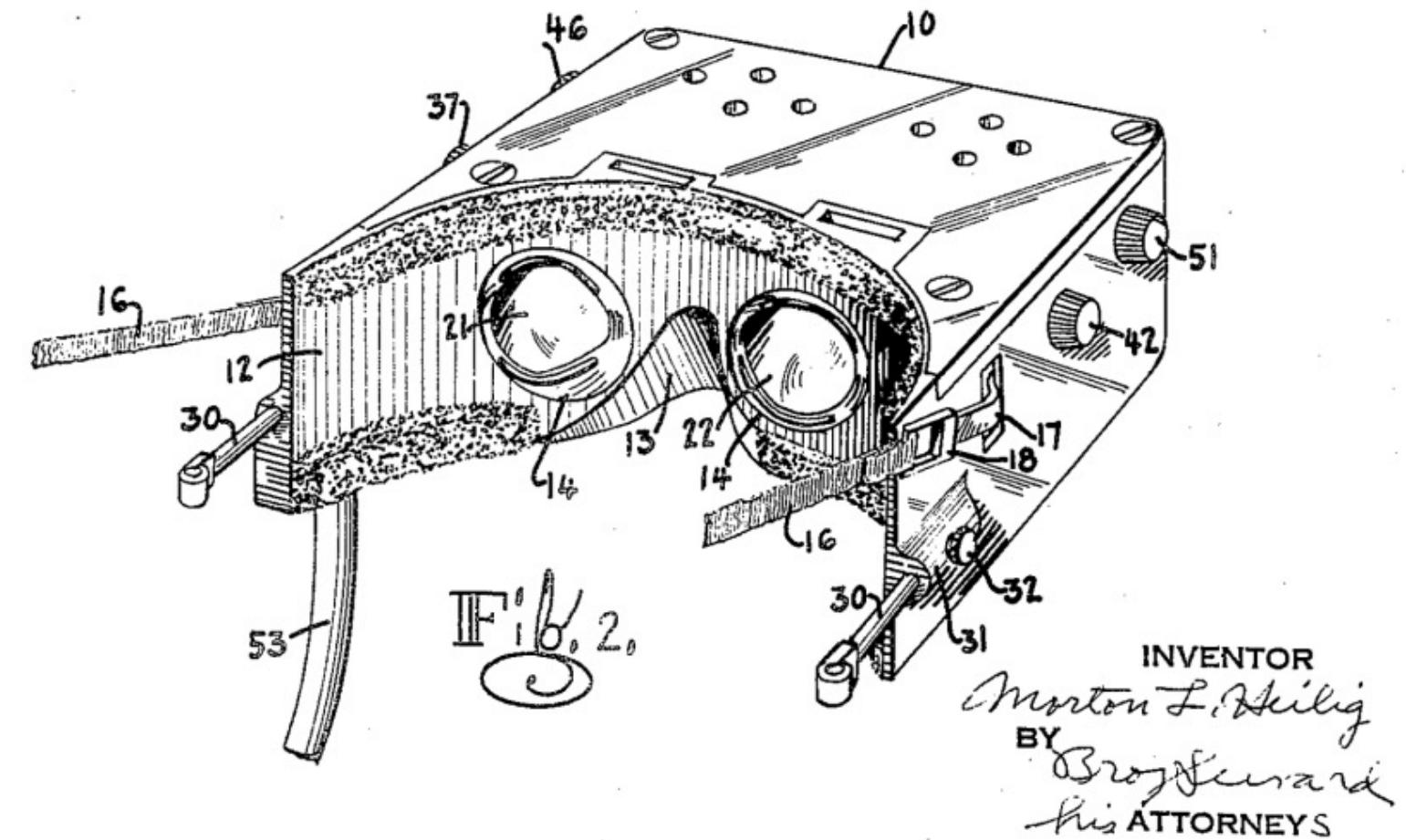


Head-mounted targeting and gun-firing interface: 1916

Sejarah VR



Flight Simulator 1928



1960 Stereoscopic Television Apparatus

Sejarah VR

Feb. 27, 1962

R. SEIBEL ETAL
COMMUNICATION DEVICE

3,022,878

Filed Jan. 11, 1960

7 Sheets-Sheet 1

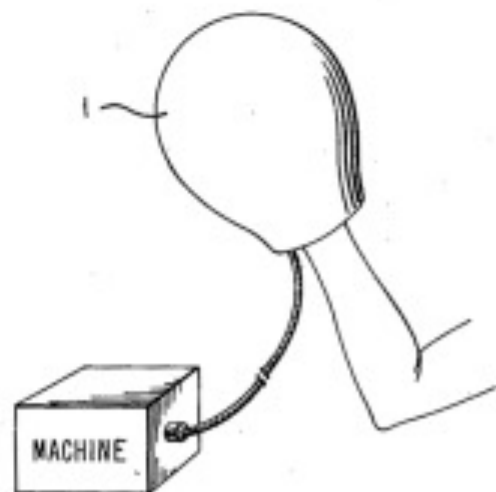


FIG. 2



FIG. 1

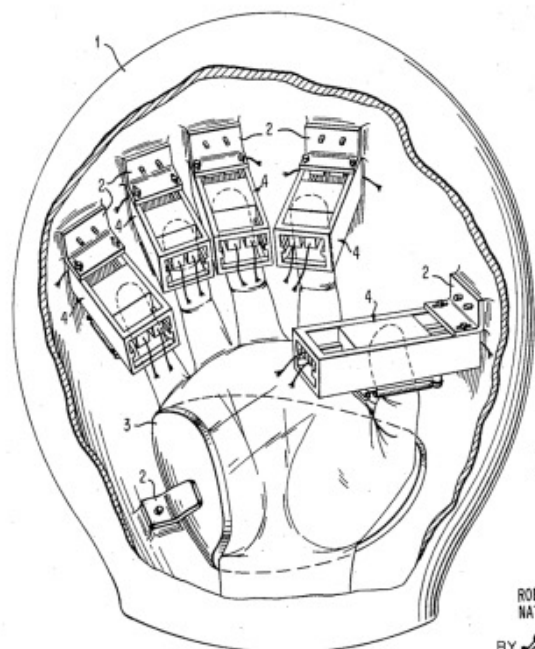
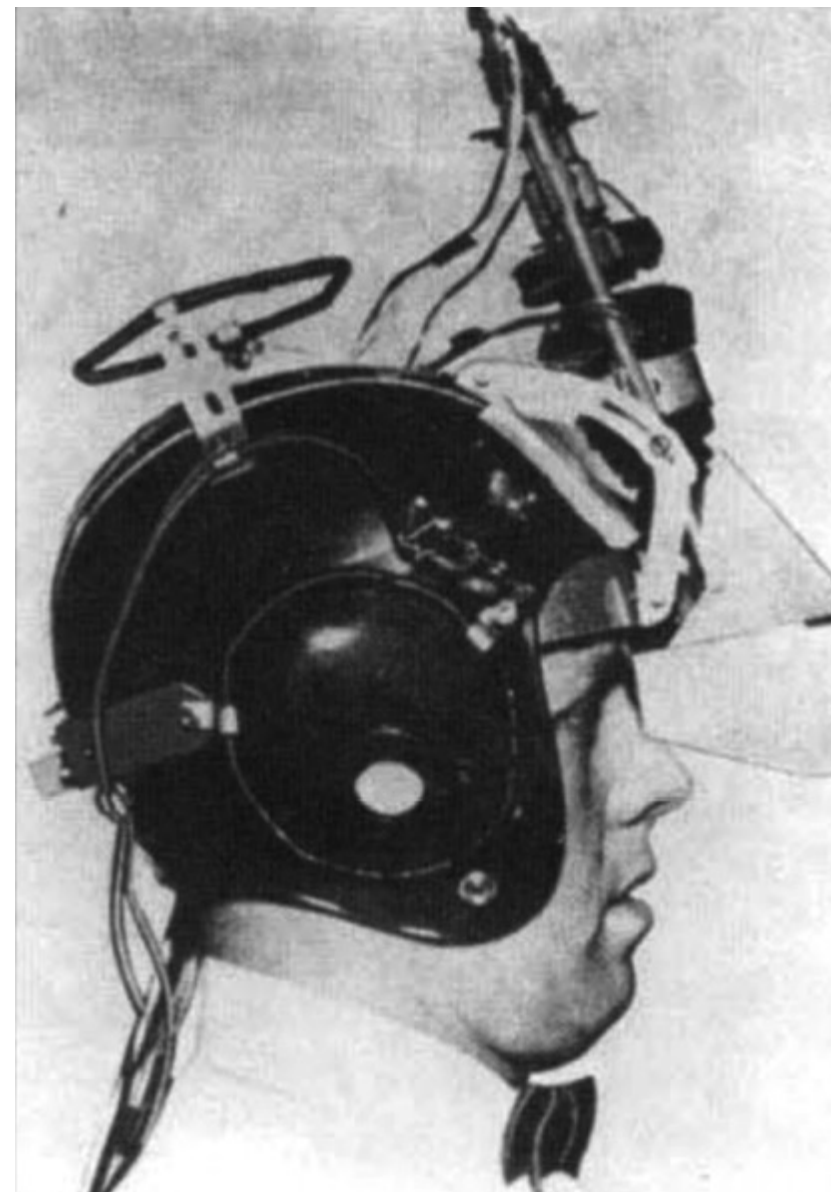


FIG. 3

INVENTORS
ROBERT SEIBEL
NATHANIEL ROCHESTER
BY *Geoffrey Knight*
ATTORNEY



Introducing . . .

sensorama

The Revolutionary Motion Picture System
that takes you into another world
with

- 3-D
- WIDE VISION
- MOTION
- COLOR
- STEREO-SOUND
- AROMAS
- WIND
- VIBRATIONS



© PATENTED

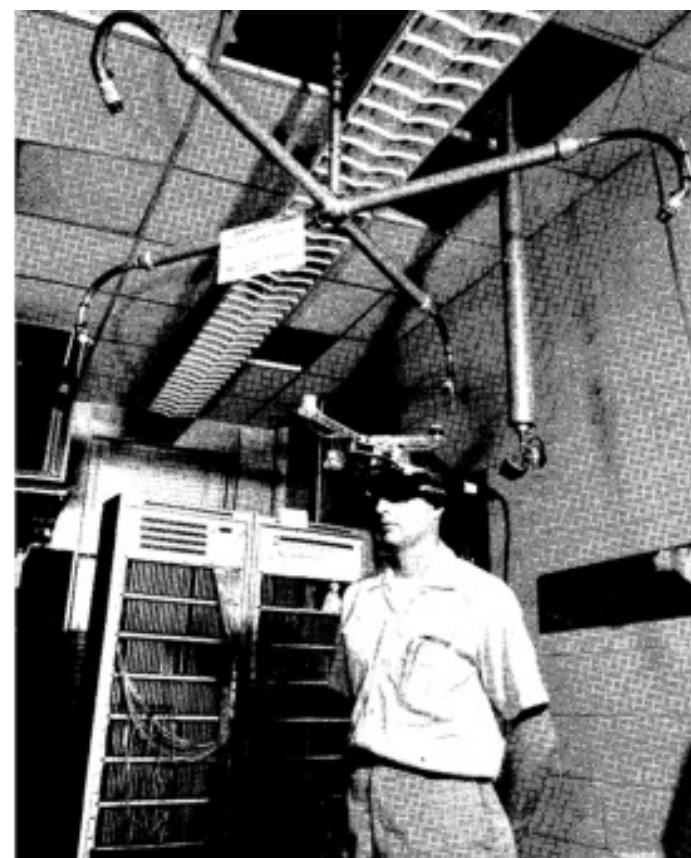
SENSORAMA, INC., 855 GALLOWAY ST., PACIFIC PALISADES, CALIF. 90272

TEL. (213) 459-2162

Sejarah VR

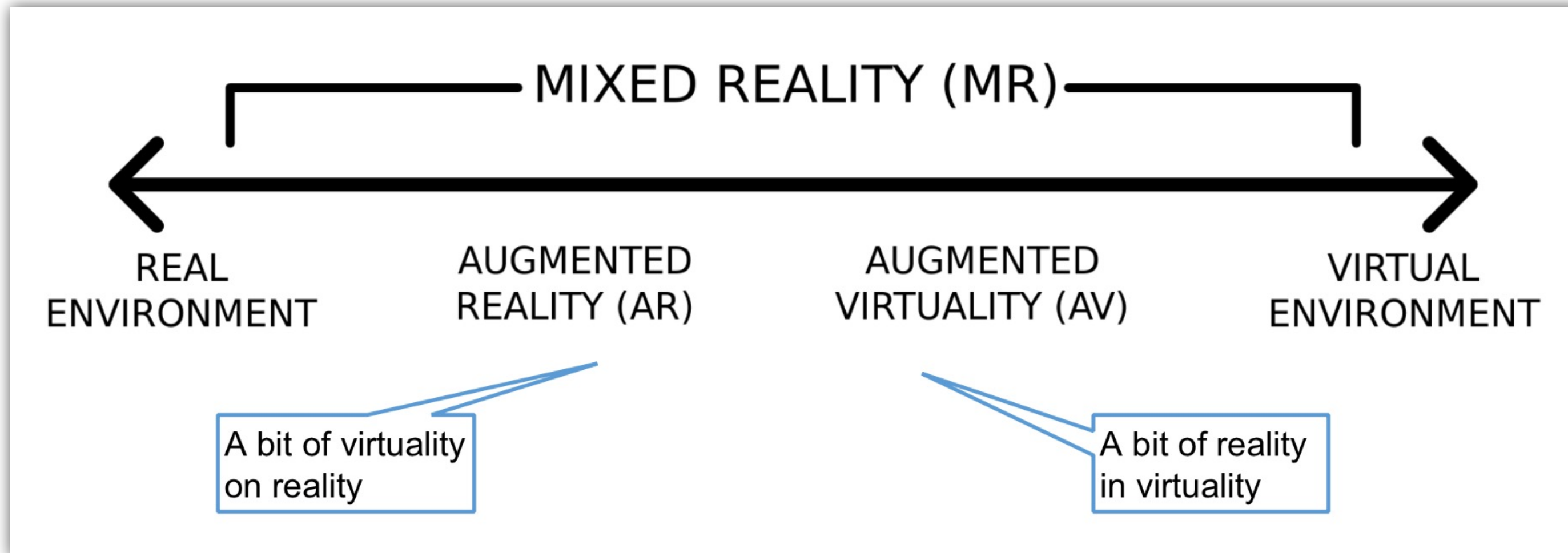


Air Force Base head-mounted display from 1967



The NASA VIEW System 1988

Kontinum Realitas Virtualitas Milgram



Indra Manusia



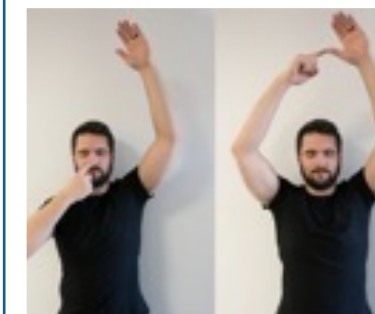
Penglihatan



Pendengaran



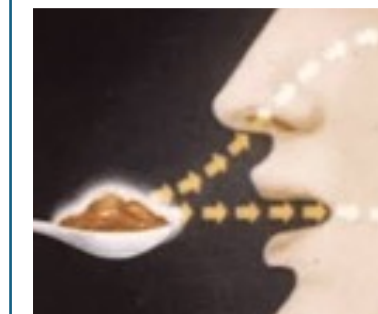
Sentuhan



Proprioception



Keseimbangan
atas Gerakan Fisik



Pembauan dan
Pengecapan

Menekan tombol

Menatapkan mata

Membuat gerakan

Berjalan

Berbicara

Emerging VR Conventions

Curved Screens

Diegetic UI First

Long Press Button

Comfort Zone

The Throwaway –
delete-gesture

Stereoscopic
Consistency

VR Smart Watch
Menu

Hand Interaction

Normal Maps

Locomotion, Snap
Rotation, Motion
Sickness Option

Constant Velocity

VR Onboarding

Text Readability

VR Button
Convention

Selection Ray
Function

HUD Motion Lag

VR Menus

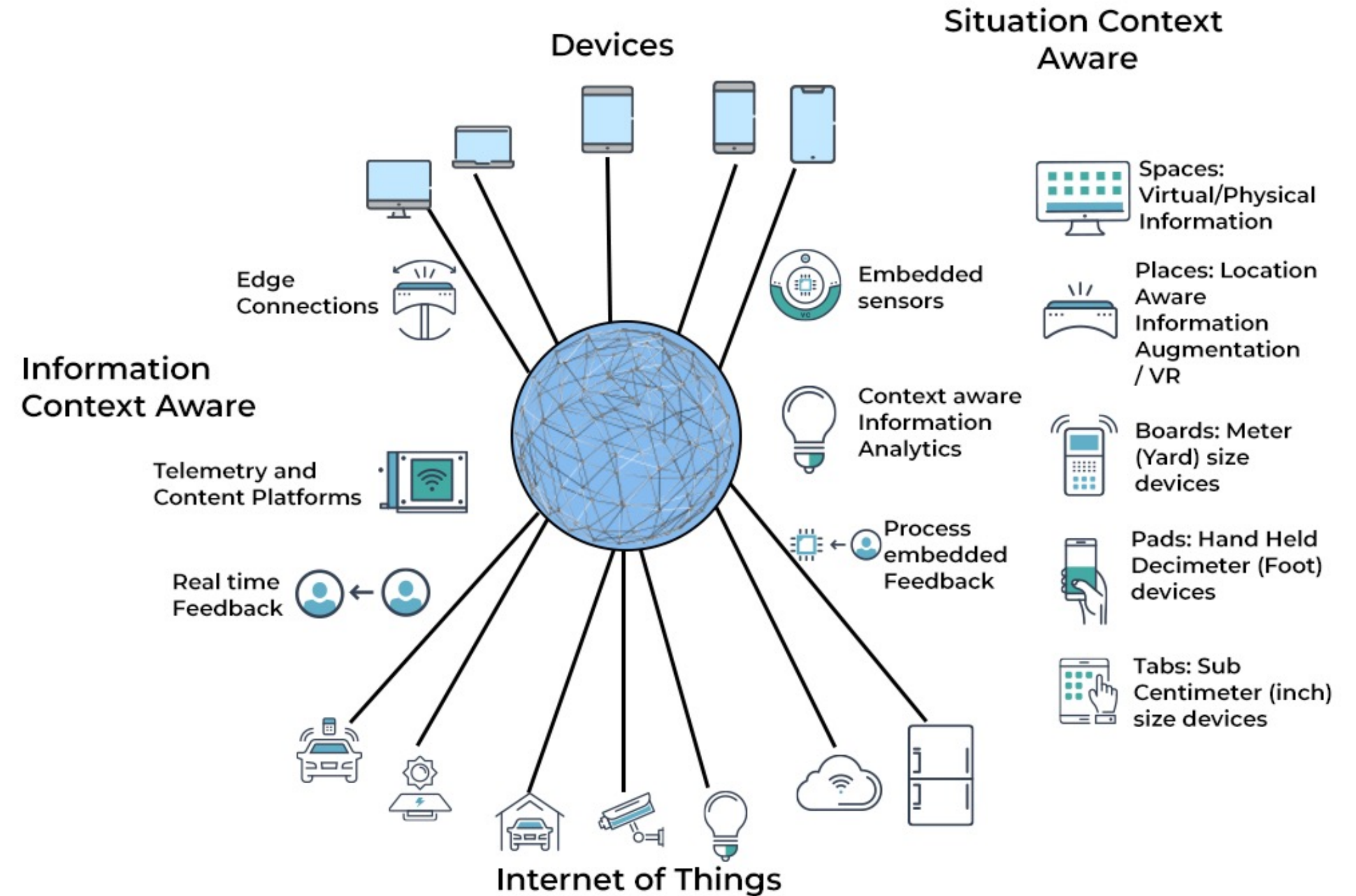
Legacy Interaction
– Gaze Activation

Augmented Reality



UbiCom & Wearables

HOW UBIQUITOUS COMPUTING WORKS



Robotics

Human Robot Interaction



Mobile

Context-aware computing

Ubiquitous computing

Augmented Reality





Kampus
Merdeka
INDONESIA JAYA



DEPARTEMEN
TEKNIK INFORMATIKA

Institut Teknologi Sepuluh Nopember

Idea

The way people interact and interface

Tim RMK GIGA
Departemen Teknik Informatika



Institut Teknologi Sepuluh Nopember



ITS_campus

its.ac.id

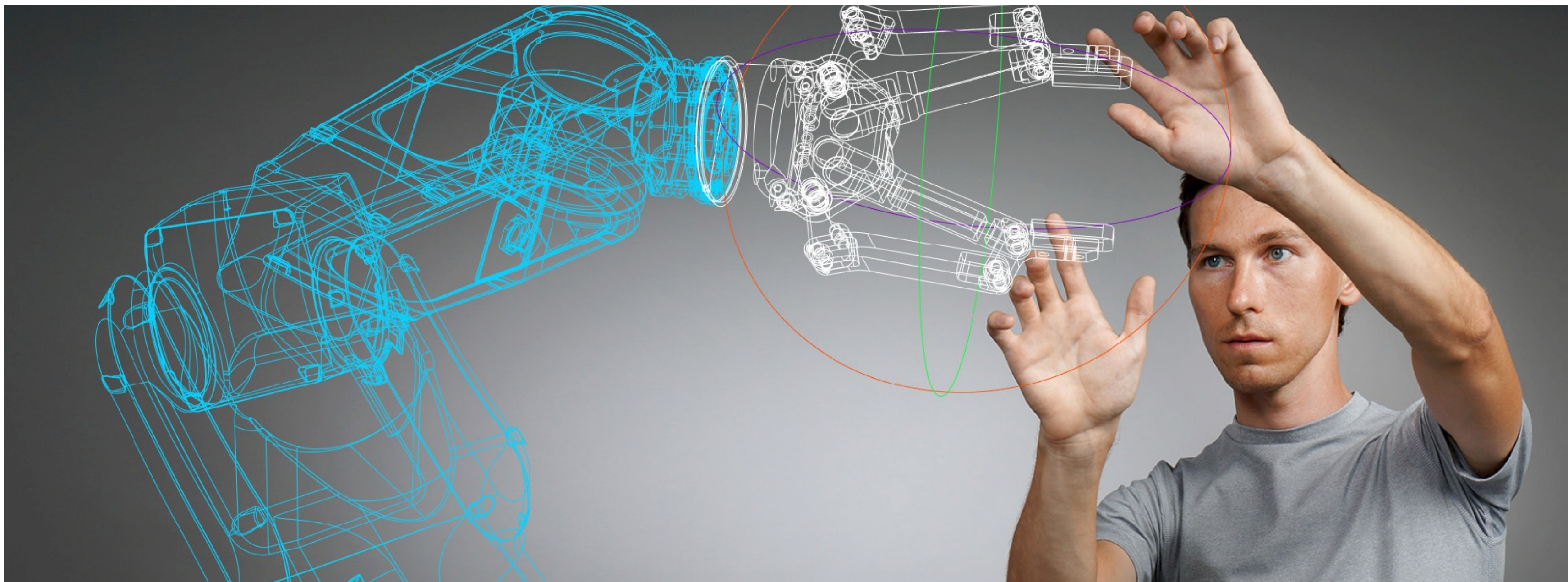
Context Sensitive Computing

Context is a fundamental part of the way humans interact with other humans

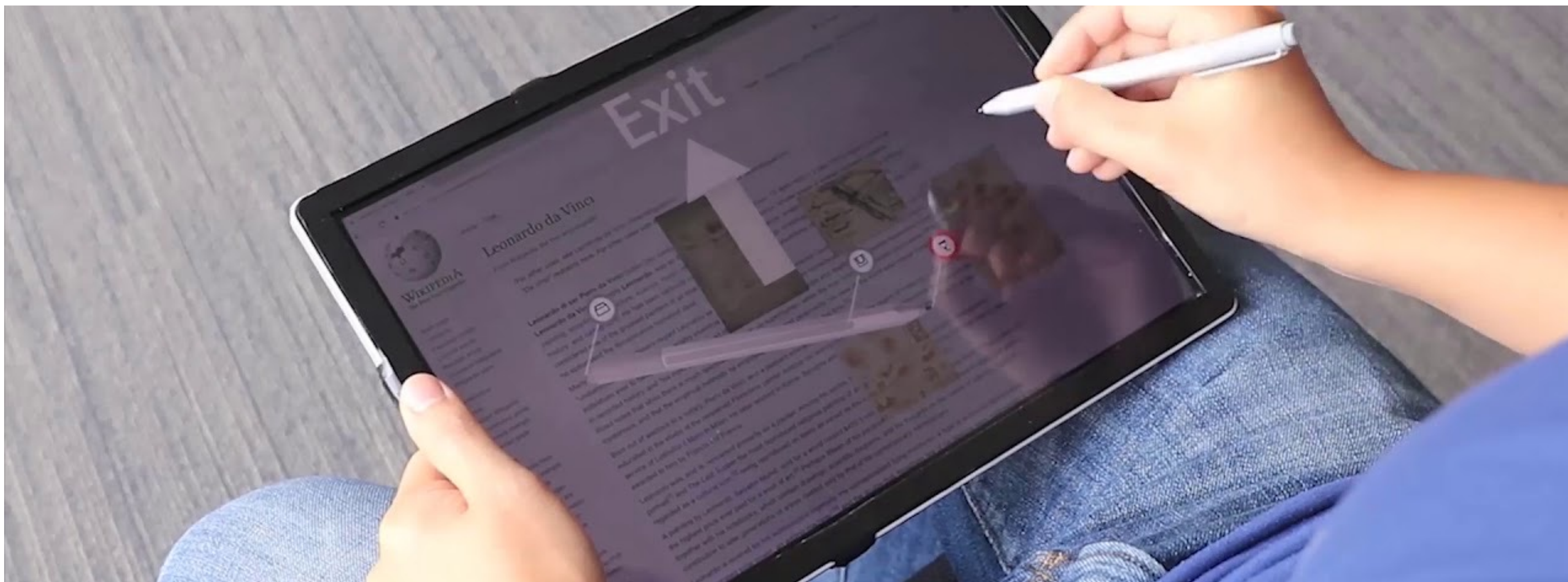
Computer must be equipped with some understanding context



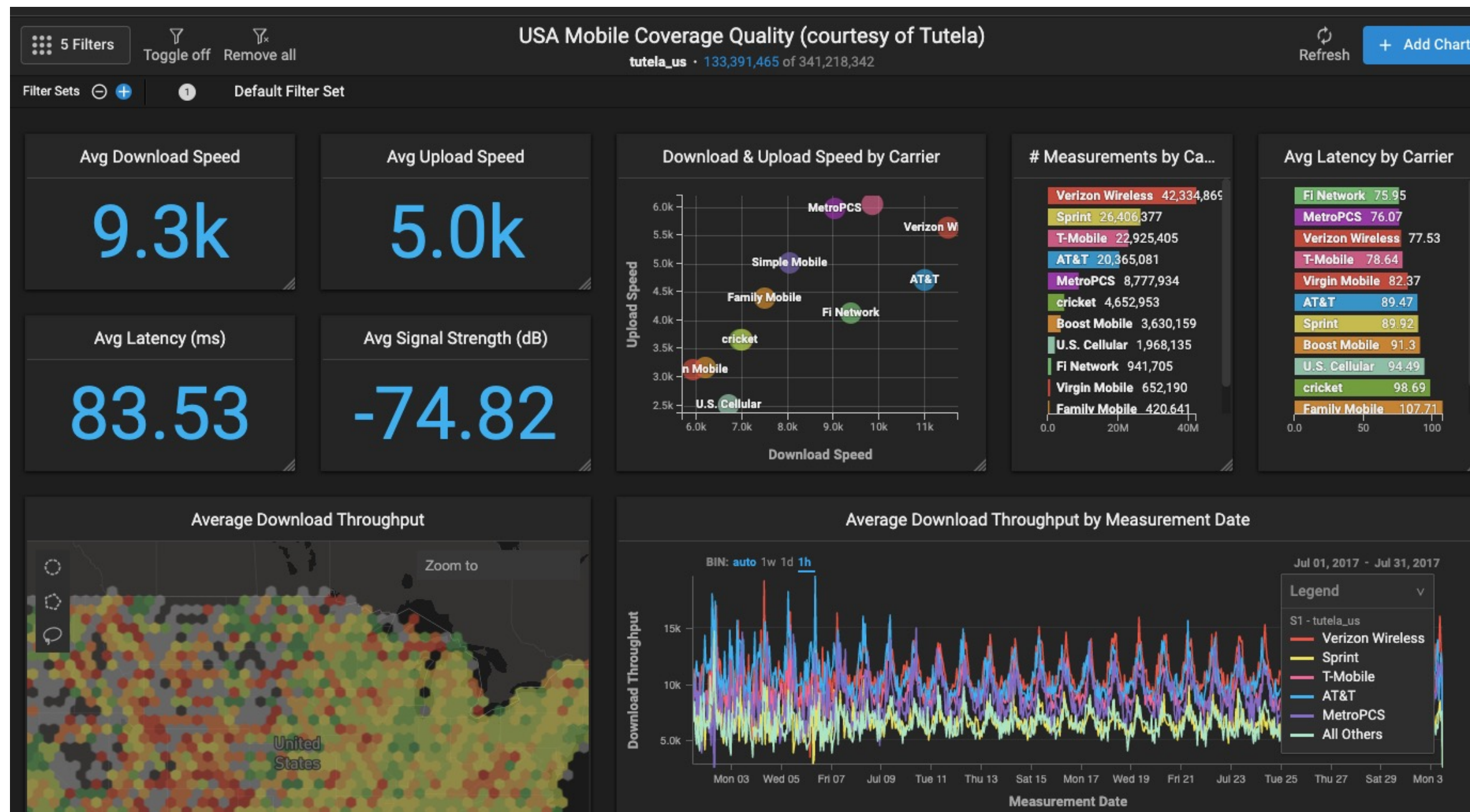
Gesture Based Interaction



Pen and Touch-based Interaction



Information Visualization





DEPARTEMEN
TEKNIK INFORMATIKA

Institut Teknologi Sepuluh Nopember

Domain



Institut Teknologi Sepuluh Nopember

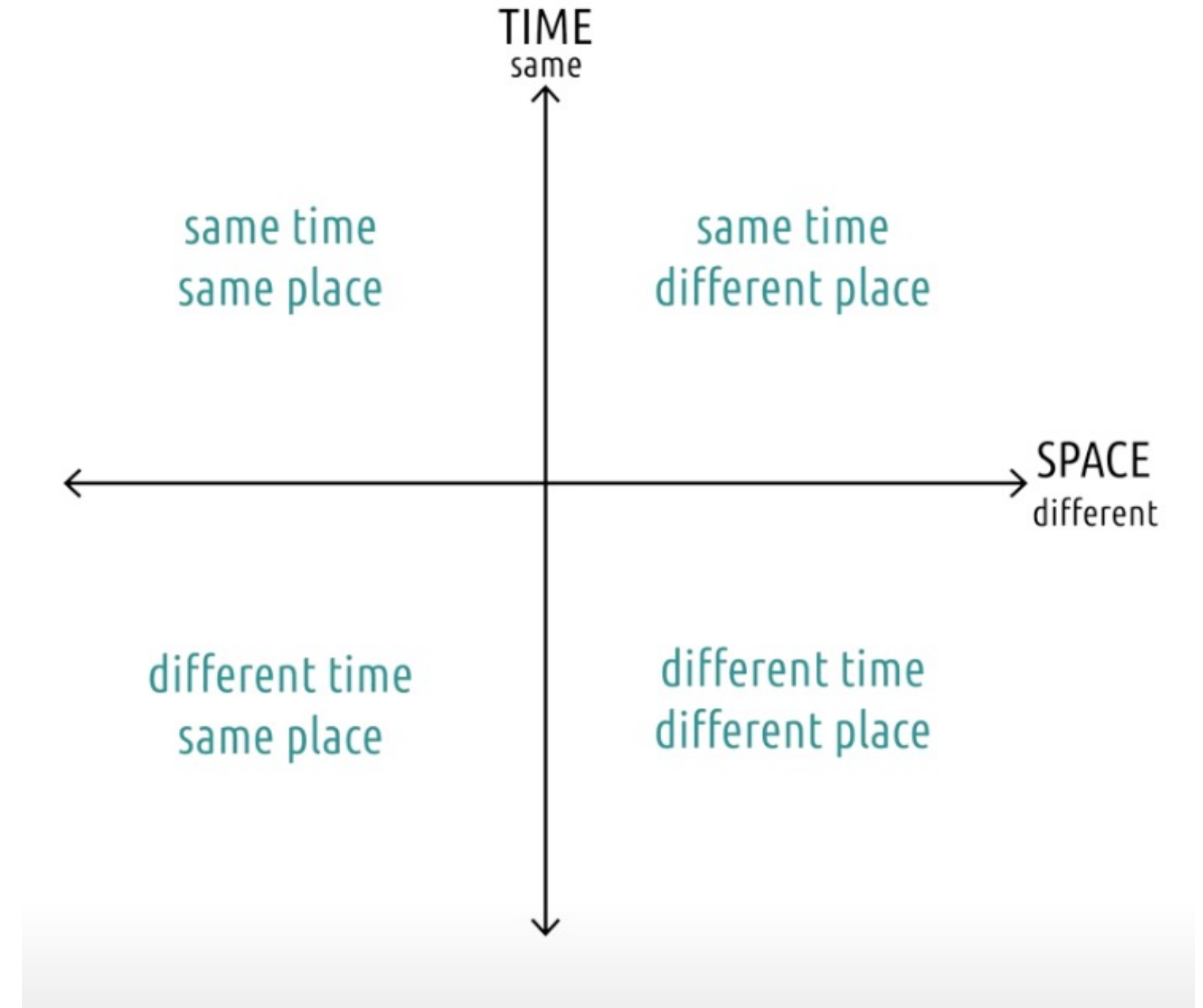


ITS_campus

its.ac.id

CSCW

CCWS or Computer-Supported Cooperative Work

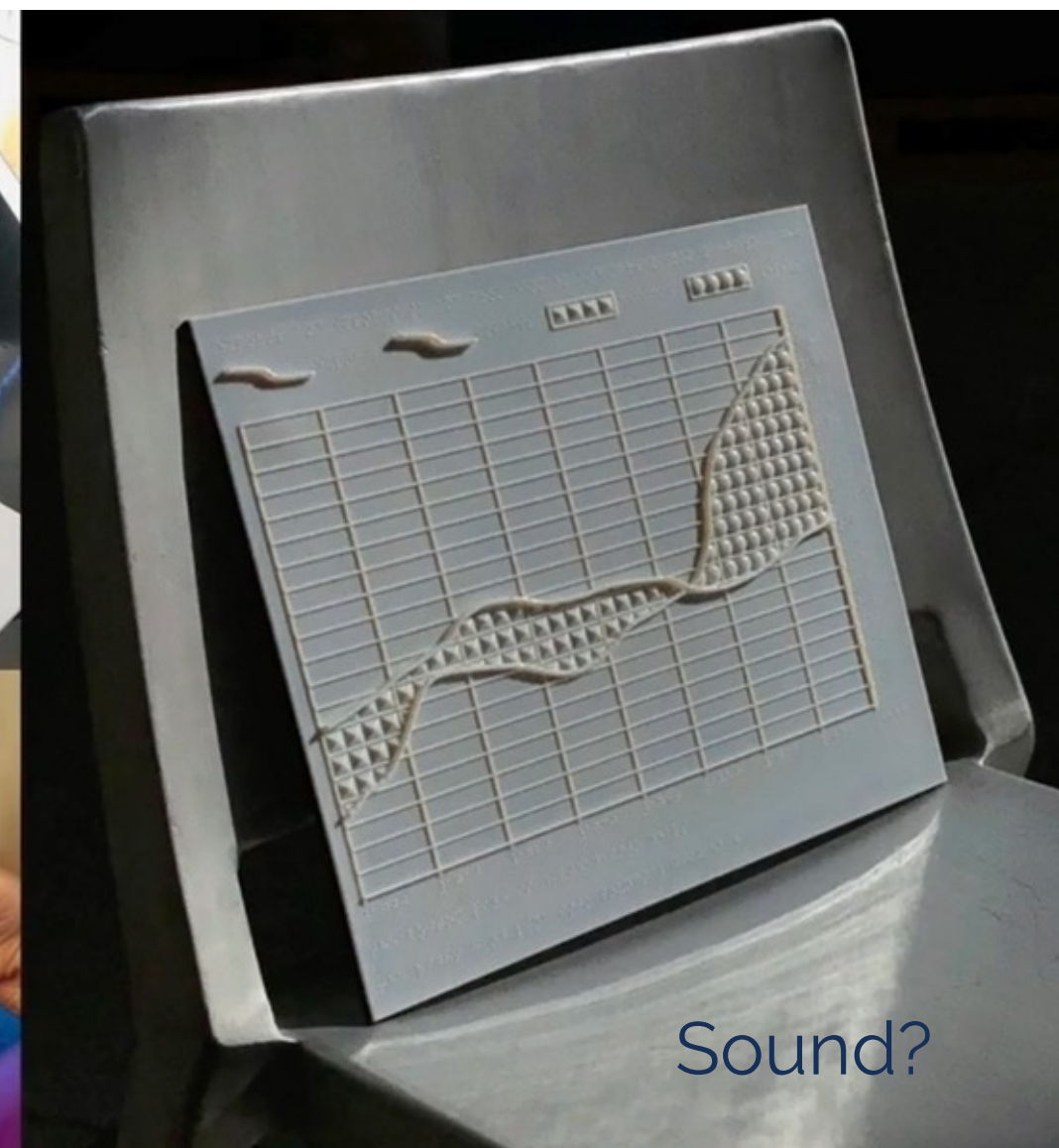
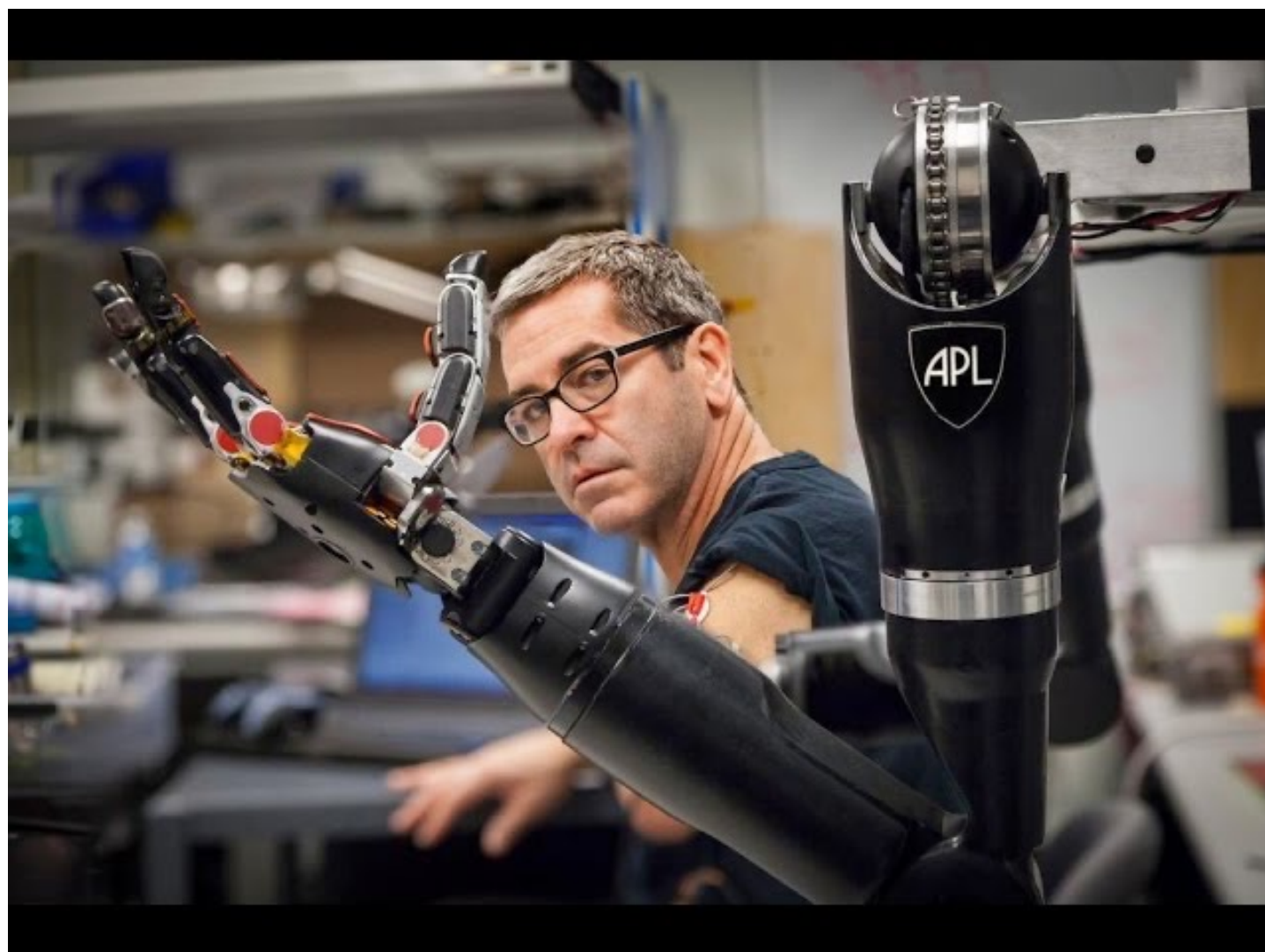


Social Computing

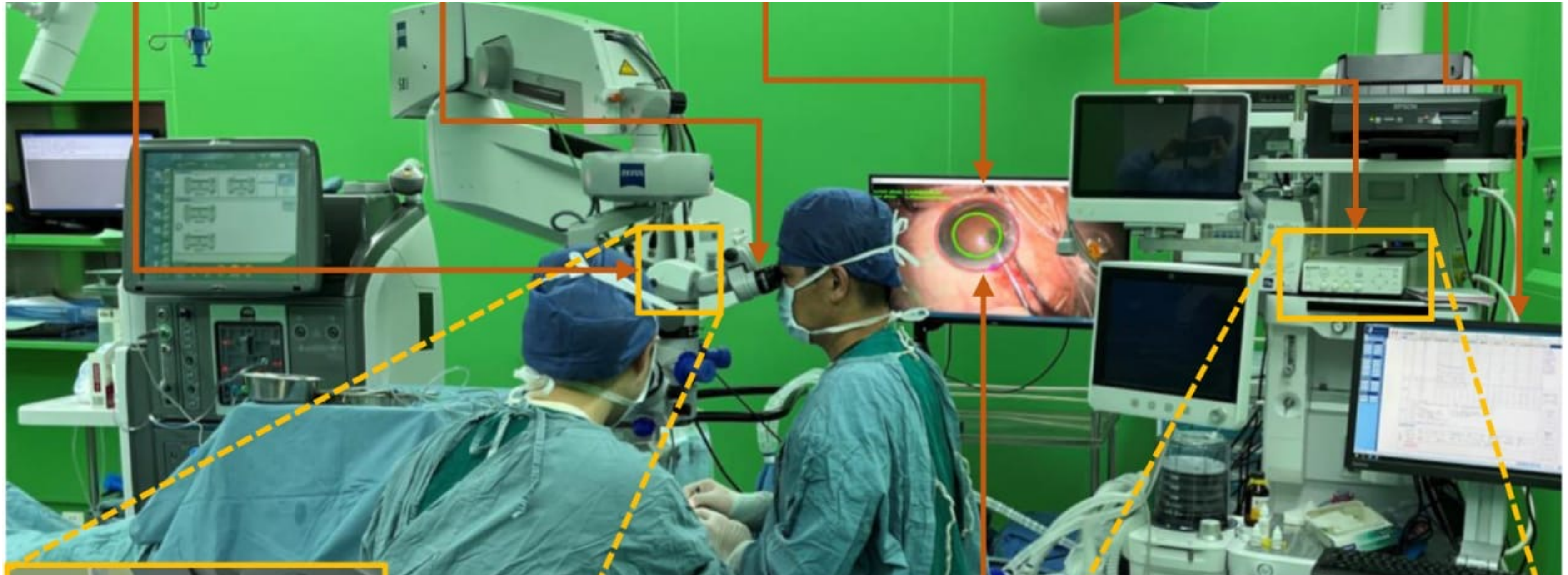
How computer affect how we interact and socialize



Special Needs



Health Care



Education



How to design HCI for education?

Security

Username

Password

How to Create a Strong Password

- ✓ Minimum 12 characters
- ✓ One capitalized letter
- One number
- At least one non letter / number (i.e. !@#\$\$%^&)

Game





Thank You

Tim RMK GIGA

